

Manzil JEE (2025)

Physics

Motion in a Straight Line

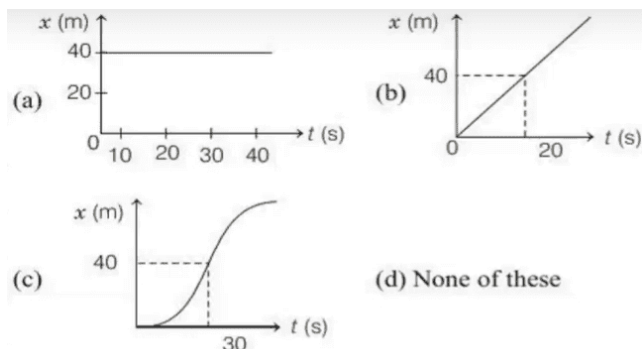
DPP: 3

- Q1** A particle is moving with speed $v = b\sqrt{x}$ along positive x -axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$)
- (A) $b^2\tau$
 (B) $\frac{b^2\tau}{4}$
 (C) $\frac{b^2\tau}{2}$
 (D) $\frac{b^2\tau}{\sqrt{2}}$
- Q2** A particle is dropped from a certain height. The time taken by it to fall through successive distances of 1 km each will be:
- (A) All equal, being equal to $\sqrt{\frac{2}{g}}$ second.
 (B) In the ratio of the square roots of the integers $1 : \sqrt{2} : \sqrt{3}$
 (C) In the ratio of the difference in the square roots of the integers, i.e., $\sqrt{1}, (\sqrt{2} - \sqrt{1}), (\sqrt{3} - \sqrt{2}), (\sqrt{4} - \sqrt{3}) \dots$
 (D) In the ratio of the reciprocals of the square roots of the integers, i.e., $\frac{1}{\sqrt{1}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \dots$
- Q3** A body starts to fall freely under gravity. The distances covered by it in first, second and third second are in ratio:
- (A) $1 : 3 : 5$ (B) $1 : 2 : 3$
 (C) $1 : 4 : 9$ (D) $1 : 5 : 6$
- Q4** A body is released from the top of a tower of height h . It takes t sec to reach the ground.

Where will be the ball after time $\frac{t}{2}$ sec

- (A) At $h/2$ from the ground
 (B) At $h/4$ from the ground
 (C) Depends upon mass and volume of the body
 (D) At $3h/4$ from the ground
- Q5** A particle moving in a straight line has velocity displacement equation as $v = 5\sqrt{1+x}$. Here v is in m/sec and x in metre. Select the correct alternative
- (A) Particle is initially at rest.
 (B) Initial velocity of the particle is 5 m/sec and the particle has a constant acceleration of 12.5 m/sec^2 .
 (C) Particle moves with uniform velocity.
 (D) None of these.
- Q6** The motion of a body falling from rest in a viscous medium is described by $\frac{dv}{dt} = A - Bv$, where A and B are constants. The velocity at time t is given by:
- (A) $\frac{A}{B}(1 - e^{-Bt})$
 (B) $A(1 - e^{-B^2t})$
 (C) ABe^{-t}
 (D) $AB^2(1 - t)$
- Q7** For a stationary object at $x = 40$ m, the position-time graph is





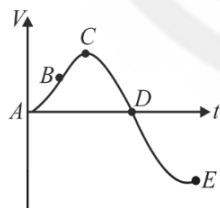
- (A) A (B) B
(C) C (D) D

Q8 Water drops fall at regular intervals from a tap which is 5 m above the ground. The third drop is leaving the tap at the instant the first drop touches the ground. How far above the ground is the second drop at that instant? (Take

$$g = 10 \text{ ms}^{-2})$$

- (A) $\frac{5}{4}$ m
(B) 4 m
(C) $\frac{5}{2}$ m
(D) $\frac{15}{4}$ m

Q9 The velocity-time graph of a particle moving along X -axis is shown in figure. Match the entries of Column I with entries of Column II



Column-I		Column-II	
I.	For AB, particle is	P.	Moving in +ve X-direction with an increasing speed
II.	For BC, particle is	Q.	Moving in +ve X-direction with a decreasing speed
III.	For CD, particle is	R.	Moving in -ve X-direction with an increasing speed
IV.	For DE, particle is	S.	Moving in -ve X-direction with a decreasing speed

- (A) $(I \rightarrow P, R), (II \rightarrow Q), (III \rightarrow S), (IV \rightarrow P)$

- (B) $(I \rightarrow P), (II \rightarrow P), (III \rightarrow Q), (IV \rightarrow R)$
(C) $(I \rightarrow R), (II \rightarrow Q), (III \rightarrow P), (IV \rightarrow S)$
(D) $(I \rightarrow Q), (II \rightarrow R), (III \rightarrow P), (IV \rightarrow S)$

Q10 Match the list-I with list-II (symbols have usual meanings)

List-I		List-II	
I	Average speed	P	$\frac{\int \vec{v} dt}{\int dt}$
II	$\frac{d^2 \vec{r}}{dt^2}$ of a particle moving in a curve	Q	$\frac{\int \vec{v} dt}{\int dt}$
III	Tangential acceleration	R	Rate of change in velocity
IV	Average velocity	S	Rate of change in speed
		T	Rate of change of displacement

- (A) I-Q II-R III-P IV-S
(B) I-Q II-R III-S IV-P
(C) I-P II-Q III-R IV-T
(D) I-P II-R III-T IV-Q

Q11 The accelerations of a particle as seen from two frames S_1 and S_2 have equal magnitude 4 m/s^2 .

- (A) The frames must be at rest with respect to each other.
(B) The frames may be moving with respect to each other but neither should be accelerated with respect to the other.
(C) The acceleration of S_2 with respect to S_1 may either be zero or 8 m/s^2 .
(D) The acceleration of S_2 with respect to S_1 may be anything between zero and 8 m/s^2 .

Q12 A scooter starting from rest moves with a constant acceleration for a time Δt_1 , then with



a constant velocity for the next Δt_2 and finally with a constant deceleration for the next Δt_3 to come to rest. A 500 N man sitting on the scooter behind the driver manages to stay at rest with respect to the scooter without touching any other part. The force exerted by the seat on the man is-

- (A) 500 N throughout the journey
- (B) less than 500 N throughout the journey
- (C) more than 500 N throughout the journey
- (D) > 500 N for time Δt_1 and Δt_3 and 500 N for Δt_2 .

Q13 From a tower of height H , a particle is thrown vertically upwards with a speed u . The time taken by the particle to hit the ground is n times that taken by it to reach the highest point of its path. The relation between H , u and n is given by

- (A) $gH = (n - 2)u^2$
- (B) $2gH = n^2u^2$
- (C) $gH = (n - 2)^2u^2$
- (D) $2gH = nu^2(n - 2)$

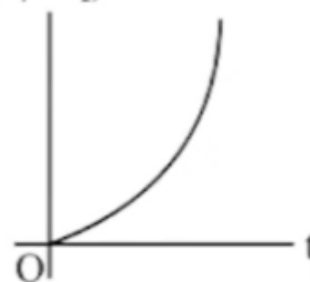
Q14 A ball projected from ground vertically upward is at same height at time t_1 and t_2 . The speed of projection of ball is [Neglect the effect of air resistance]

- (A) $g[t_2 - t_1]$
- (B) $\frac{g[t_1 + t_2]}{2}$
- (C) $\frac{g[t_2 - t_1]}{2}$
- (D) $g[t_1 + t_2]$

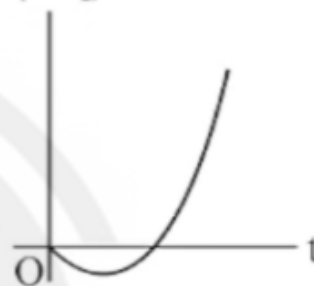
Q15 A body is at rest at $x = 0$. At $t = 0$, it starts moving in the positive x -direction with a constant acceleration. At the same instant another body passes through $x = 0$ moving in the positive x -direction with a constant speed. The position of the first body is given by $x_1(t)$

after t and that of the second body by $x_2(t)$ after the same time interval. Which of the following graphs correctly describes $(x_1 - x_2)$ as a function of time t ?

(A) $(x_1 - x_2)$



(B) $(x_1 - x_2)$



(C) $(x_1 - x_2)$



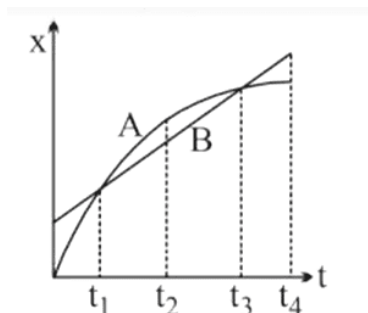
(D) $(x_1 - x_2)$



Q16 The graph given shows the POSITION of two cars, A and B, as a function of time. The cars move along the x -axis on parallel but separate



tracks so that they can pass each other's position without colliding.



At which instant do the two cars have the same velocity?

- (A) t_1
- (B) t_2
- (C) t_3
- (D) t_4

Q17 A ball is bouncing elastically with a speed 1 m/s between walls of a railways compartment of size 10 m in a direction perpendicular to walls. The train is moving at 10 m/s along the direction of motion of ball. As seen from the ground which statement is incorrect.

- (A) The direction of motion of ball changes every 10 seconds
- (B) speed of ball changes every 10 seconds
- (C) Average speed of ball over any 20 seconds interval is fixed
- (D) The acceleration is the same as seen from the train

Q18 A particle starting from rest with constant acceleration travels a distance x in first 2s and a distance y in next 2 s, then

- (A) $y = x$
- (B) $y = 2x$
- (C) $y = 3x$
- (D) $y = 4x$

Q19 A ball is thrown vertically upwards from the top of a tower of height h with velocity v . The ball strikes the ground after time.

- (A) $\frac{v}{g} \left[1 + \sqrt{1 + \frac{2gh}{v^2}} \right]$
- (B) $\frac{v}{g} \left[1 - \sqrt{1 + \frac{2gh}{v^2}} \right]$
- (C) $\frac{v}{g} \left(1 + \frac{2gh}{v^2} \right)^{1/2}$
- (D) $\frac{v}{g} \left(1 - \frac{2gh}{v^2} \right)^{1/2}$

Q20 A particle of unit mass undergoes one dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$ where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x , is given by:

- (A) $-2n\beta^2 x^{-4n-1}$
- (B) $-2n\beta^2 e^{-4n+1}$
- (C) $-2\beta^2 x^{-2n+1}$
- (D) $-2n\beta^2 x^{-2n-1}$

Q21 A man is going up in a lift (open at the top) moving with a constant velocity 3 m/s. He throws a ball up at 5 m/sec relative to the lift when the lift is 50 m above the ground. Height of the lift when the ball meets it during its downward journey is ($g = 10 \text{ m/s}^2$)

- (A) 53 m
- (B) 58 m
- (C) 56 m
- (D) 68 m

Q22 A particle moves along a straight line such that its displacement S varies with time t as $S = \alpha + \beta t + \gamma t^2$



List I		List II	
I	Acceleration at $t = 2$ s	P	$\beta + 5\gamma$
II	Average velocity during 3 rd s	Q	2γ
III	Velocity at $t = 1$ s	R	α
IV	Initial displacement	S	$\beta + 2\gamma$

- (A) I-P, II-Q, III-S, IV-R
 (B) I-Q, II-P, III-R, IV-Q
 (C) I-P, II-Q, III-R, IV-S
 (D) I-Q, II-P, III-S, IV-R

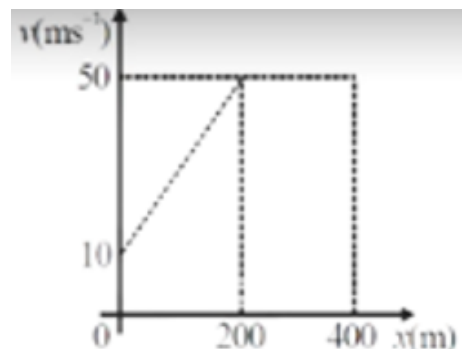
Q23 A point object traverses half the distance with velocity v_0 . The remaining part of the distance is covered with velocity v_1 for the half time and with velocity v_2 for the rest half. The average velocity of the object for the whole journey is

- (A) $\frac{2v_1(v_0+v_2)}{(v_0+2v_1+2v_2)}$
 (B) $\frac{2v(v_0+v_1)}{(v_0+v_1+v_2)}$
 (C) $\frac{2v_0(v_1+v_2)}{(v_1+v_2+2v_0)}$
 (D) $\frac{2v_2(v_0+v_1)}{(v_1+2v_2+v_0)}$

Q24 A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h . Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively

- (A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$
 (B) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$
 (C) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$
 (D) $\frac{1}{3}$

Q25 The velocity-displacement graph describing the motion of a bicycle is shown in the figure.



The acceleration-displacement graph of the bicycle's motion is best described by

- (A)
 (B)
 (C)
 (D)



Answer Key

Q1 C
Q2 C
Q3 A
Q4 D
Q5 B
Q6 A
Q7 A
Q8 D
Q9 B
Q10 B
Q11 D
Q12 D
Q13 D

Q14 B
Q15 B
Q16 B
Q17 A
Q18 C
Q19 A
Q20 A
Q21 A
Q22 D
Q23 C
Q24 B
Q25 A



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